Here we see some other examples of integrals where partial fraction decomposition is useful.

1. Evaluate $\int \ln(x^2-4)dx$.

2. Evaluate each of the following integrals by making a substitution which turns the integrand into a rational function.

   (a) $\displaystyle\int \frac{\cos(x)}{\sin^2(x)+\sin(x)}dx$

(b) $\int \frac{e^{2x}}{e^{2x}+3e^{x}+2}dx$

(c) $\int \frac{\sqrt{x+4}}{x} dx$